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Subject Code: MPH103T

Roll No:

MPHARM**(SEM. I) THEORY EXAMINATION 2025-26****MODERN PHARMACEUTICS****TIME: 3 HRS****M.MARKS: 75****Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2x10 = 20**

(a)	Define preformulation studies.
(b)	What are drug–excipient interactions?
(c)	Define stability testing.
(d)	What is SMEDDS?
(e)	Define optimization in pharmaceutical formulation.
(f)	What is factorial design?
(g)	Define process validation.
(h)	What is cGMP?
(i)	Define Heckel plot.
(j)	What is ANOVA test?

SECTION B**2. Attempt any two of the following:****2 x 10 = 20**

(a)	Discuss different methods used for study of drug–excipient interactions and explain kinetics of stability testing.
(b)	Explain formulation, manufacturing and evaluation of emulsions and suspensions.
(c)	Describe optimization techniques in pharmaceutical formulation with special reference to factorial design and response surface methodology.

SECTION C**3. Attempt any five part of the following:****7 x 5 = 35**

(a)	Write a note on large volume parenterals.
(b)	Discuss contour designs and their applications.
(c)	Explain types of pharmaceutical validation.
(d)	Write a short note on production planning and control.
(e)	Explain physics of tablet compression.
(f)	Discuss Higuchi and Peppas plots.
(g)	Write a note on Chi-square test and Student's t-test.